

COURSE TITLE : **NETWORK PROGRAMMING**
COURSE CODE : **5152**
COURSE CATEGORY : **A**
PERIODS/WEEK : **4**
PERIODS/SEMESTER : **52**
CREDITS : **4**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Concepts of Object Oriented Programming	13
2	Multithreading, Exceptions and Applets	13
3	Uniform Resource Locator (URL) and Socket	13
4	RMI and Client-Server concepts	13

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able :
1	1	To understand the Concepts of Object Oriented Programming
	2	To understand Arrays, Inheritance ,Interfaces and packages
2	1	To understand Multithreading, Exceptions and Applets
	2	To understand Stream classes and GUI programming with Swing
3	1	To understand Uniform Resource Locator URL
	2	To understand the Client-server programming using Socket
4	1	To understand RMI and Client-Server concepts

Specific Outcomes:

MODULE I Concepts of Object Oriented Programming

- 1.1 To understand the Concepts of Object Oriented Programming
 - 1.1.1 Explain the features of Java
 - 1.1.2 Describe Java Virtual machine and Byte code
 - 1.1.3 Describe the structure of java program
 - 1.1.4 Explain the implementation of java programs
 - 1.1.5 Explain classes, objects and methods

- 1.1.6 Illustrate constructors with examples
- 1.1.7 Describe the concepts of overloading

1.2 To understand Arrays, Inheritance ,Interfaces and packages

- 1.2.1 Explain Arrays and Strings in java
- 1.2.2 Define Inheritance
- 1.2.3 Describe different types of Inheritances.
- 1.2.4 Explain Visibility controls
- 1.2.5 Describe final class and abstract classes.
- 1.2.6 Explain Interfaces and multiple inheritance.
- 1.2.7 Explain how to use System packages
- 1.2.8 Discuss Naming conventions
- 1.2.9 Describe the steps to create packages
- 1.2.10** Describe how packages can be accessed.

MODULE II Multithreading, Exceptions and Applets

1.2 2.1.0 To understand Multithreading, Exceptions and Applets

- 1.1.1 Explain multithreading in java
- 1.1.2 Explain how to create a thread
- 1.1.3 Explain Life cycle of a thread
- 1.1.4 Discuss thread methods
- 1.1.5 Discuss thread priority
- 1.1.6 Explain Synchronisation
- 1.1.7 Discuss Exception handling mechanism in java
- 2.1.8 Discuss Multiple catch statements and finally statements
- 2.1.9 Explain Applet programming

2.2.0 To understand Stream classes and GUI programming with Swing

- 2.2.1 Discuss Concepts of streams
- 1.2.2 Discuss Byte Stream and character stream classes
- 2.2.3 Explain how a data file can be created in java
- 2.2.4 Describe Reading/Writing bytes in files
- 2.5.1 Describe Reading/writing characters in files.
- 2.5.2 Discuss Swing Classes and GUI Components
- 2.2.7 Discuss Event handling in java

MODULE III Uniform Resource Locator (URL) and Socket

1.1.0 To understand Uniform Resource Locator URL

- 1.1.1 State definition and format of URL
- 1.1.2 Identify the components
- 1.1.3 Discuss Java URL Object and its use
- 1.1.4 Discuss about absolute and relative URL
- 3.1.5 Describe Various URL constructors - open stream and open connection methods

1.2 To understand the Client-server programming using Socket

- 1.2.0 Discuss ports and port numbers
- 1.2.1 Discuss about concept of the Socket
- 1.2.2 Describe different types of Sockets such as TCP and UDP
- 1.2.3 Discuss creation of TCP Sockets
- 3.2.5 Distinguish between Server side program and Client side program with suitable examples
- 1.2.4 Describe creation procedure for server side programming.
- 1.2.5 Discuss about Socket class and Server Socket class.
- 1.2.6 Describe creation procedure for client side programming
- 1.2.7 Explain how to support of multiple clients using java
- 1.2.8 Discuss the creation of client server application using UDP

MODULE IV RMI and Client-Server concepts

- 4.1.0 To understand RMI and Client-Server concepts
 - 4.1.1 State Remote Method Invocation
- 4.1.2 Discuss the steps for developing an RMI system
- 4.1.3 Discuss Remote class and Interfaces in java
- 4.1.4 Explain how to Program a Client in Java
- 4.1.5 Explain how to Program a Server in java
- 4.1.6 Describe the procedure to start the Server in java
- 4.1.7 Describe how to Run a Client in java
- 4.1.8 Discuss Security in java

CONTENT DETAILS

Module I Concepts of Object Oriented Programming

Brief History of Java , Feature of Java , JDK , Data Types , Operators ,Control Structures in JAVA , Arrays , The JAVA Class , Constructor , Finalizers , Classes inside Classes . Inheritance & Interface , Deriving Classes ,Method Overriding , Method Overloading , Access Modifiers , Abstract Class and Method , Interfaces , Packages.

Module II Multithreading, Exceptions and Applets

Exception Handling , The Try-Catch Statement ,Catching more than one Exception , The Finally Clause , When to use Exceptions . Threads: Introduction , Creating Treads in Applications ,Methods in Thread Class .

Java I/O Packages , Java Input Stream Classes , Java Output ,File Class .Graphic & Sound: AWT and Swing , Graphic Methods , Fonts , User interface components with Swing ,Loading and Viewing Images ,Loading and Playing Sound , AWT & Event Handling ,Layout.

MODULE III Uniform Resource Locator (URL) and Socket

Uniform Resource Locator (URL)-definition and format, components, URL objects and use, absolute and relative URL, URL constructors.

Socket-concepts, port and port numbers, types of sockets-TCP and UDP ,TCP socket creation, Client-side and Server-side programming, socket class, multiple clients in java, UDP-client server application

MODULE IV RMI and Client-Server concepts

RMI-definition, steps for development, remote class and interface, client-side program, server-side program, starting the server, running the client, Security

Text Book(s):-

- 4.1 Java Network Programming– E R Harold , O'Reilly Media, Second Edition
- 4.2 Java Fundamentals-by Herbert Schildt, Dale Skrien-Tata Mc Graw-Hill –First Edition.

REFERENCE:-

1. The Java Programming Language- Ken Arnold, James Gosling- Addison-Wesley- Second Edition.
2. Java Programming Advanced Topics- Joe Wigglesworth, Paula Lumby-Thomson Learning-1998
3. Java Network Programming- Elliotte Rusty Harold-O'Reilly Media-3RD Edition